

GE Healthcare

Signa HDx Series VRE Recon Upgrade

Technical Publication



5177003
Rev 6

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation “**damage in shipment**” written on **all** copies of the freight or express bill **before** delivery is accepted or “signed for” by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

File a report with

- Call 1–800–548–3366 and use option 8.
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

© 2006 – 2008, General Electric Company
GE Medical Systems, a General Electric Company, going to market as GE Healthcare
3002 N. Grandview Boulevard
Waukesha, Wisconsin 53149
USA

www.gehealthcare.com



Table of Contents

DESCRIPTION	PAGE
Table Of Contents	1
Language Policy For Service Documentation	3
Revision History	5
SECTION 1 – Overview	7
SECTION 2 – Upgrade Installation Process	9
SECTION 3 – Power Off	13
SECTION 4 – Upgrade Procedure	14
SECTION 5 – Completion	26

This page left blank intentionally.

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

警告

- ・ このサービスマニュアルには英語版しかありません。
- ・ GEMS以外でサービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- ・ このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないで下さい。
- ・ この警告に従わない場合、サービスを担当される方、操作員あるいは患者さんが、感電や機械的又はその他の危険により負傷する可能性があります。

注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

Revision History

Date	Rev	Change Description
2-14-2006	1	Initial Release
6-30-2006	2	Corrected spelling and cable part numbers on cable maps. Clarified the power cable connections to the Infiniband Switch. Added existing cables from Term Server to MGD Chassis. (MRIhc16177, MRIhc16207)
10-30-2006	3	Section 1: Removed Infiniband Switch from list of components added for this upgrade. Section 2: Updated flowchart to remove Infinibnad Switch reference. Section 4: Design change required removal of Infiniband Switch installation. Update cable maps and cabinet upgrade instructions. Added note to remind FE to turn on circuit breakers when upgrade is complete (MRIhc16206).
12-4-2006	4	Section 1: Removed Infiniband Switch from list of components added for this upgrade (MRIhc20582)
2-22-2007	5	Section 4: Corrected terminal strip part number. Revised placement of ICN modules. Added RRF chassis power cord connection to cable map for M3335RB.
9-18-2008	6	Changed title of manual to <i>Signa HDx Series VRE Recon Upgrade</i> . Section 1: Revised catalog offerings to reflect ICN changes. Section 2: Revised flowcharts

LIST OF EFFECTIVE PAGES

PAGE REV PAGE REV PAGE REV PAGE REV PAGE REV

Title Page 6

1 to 27 6

This page left blank intentionally.

SECTION 1 – OVERVIEW

1-1 INTRODUCTION

This manual provides instructions to upgrade a Signa HDx system from either:

A: A two 5127452 ICN to a two 5127452-3 ICN VRE system

This Direction covers the following catalogs:

M3335BH 2 VRE Recon Upgrade

This upgrade adds:

- Two ICN Modules and brackets
- Power Cables
- Power Outlet Strip (needed only for pre-HDx system cabinets)

OR:

B: A one 5127452-3 ICN to a two 5127452-3 ICN VRE system

This Direction covers the following catalogs:

M3335RC HDx VRE Recon upgrade

This upgrade adds:

- One ICN Module and brackets
- Power Cables
- VRE Cables
- Power outlet strip (needed only for pre-HDx system cabinets)

1-2 SITE READY CHECK

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

1-3 DISCARDED MATERIAL RETURN POLICY

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1-4 PRODUCT LOCATOR

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the *FE Site Verification Web Site* at: <http://gein2.med.ge.com/gib>

The FE Site Verification consists of three components that are available on the web from the main menu. They are:

- Install/deinstall product locator model and serial numbers
- Add/modify ship to address information
- Update CARES FE data for primary/secondary FEs

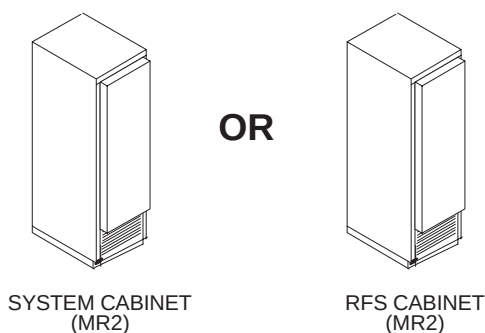
To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: [Product Locator Support Central Page](http://supportcentral.ge.com/15563) – <http://supportcentral.ge.com/15563>

2. One “Shipping Card” is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the “Installation Card” and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1-5 UPGRADE SYSTEM CONFIGURATION

Illustration 1-1 shows the major equipment modified for this upgrade.



SIGNA SERIES EXISTING HARDWARE MODIFIED FOR VRE CONFIGURATION UPGRADE
ILLUSTRATION 1-1

SECTION 2 – UPGRADE INSTALLATION PROCESS

2-1 SITE READINESS

All on-site construction must be completed before equipment is delivered and installation starts. Attempting to upgrade the system while construction is being completed will impact installation efficiency and further delay site completion. Making sure that all preinstallation and construction work is completed before equipment is delivered will usually result in an earlier turnover date.

**WARNING!**

FERROUS MATERIAL HAZARD!

DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS UPGRADE CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES. KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

2-2 UPGRADE PROCEDURE

The following upgrade flow charts contain information for upgrading to Signa HDx 3.0T Release 14.x or HDxt Release 15.x from either:

A: A two ICN (5127452) to a 2 ICN (5127452-3) VRE system.

or

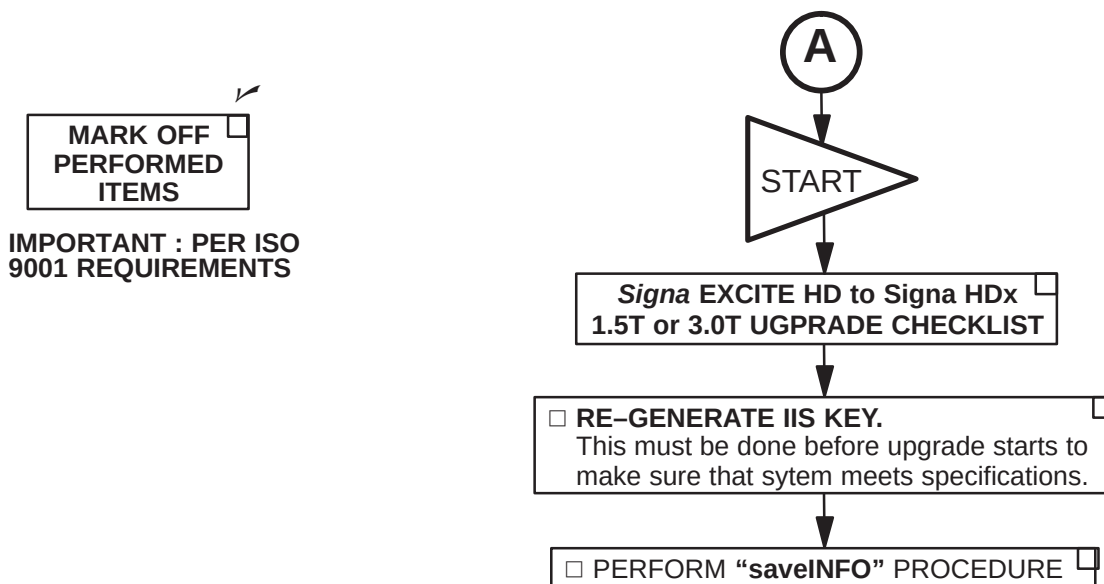
B: A one (5127452-3) to a two (5127453-3) VRE system.

The upgrade flow chart should be followed for an orderly and efficient installation. Note that procedures may be performed in parallel and may be performed in any order according to the specific situation of each site.

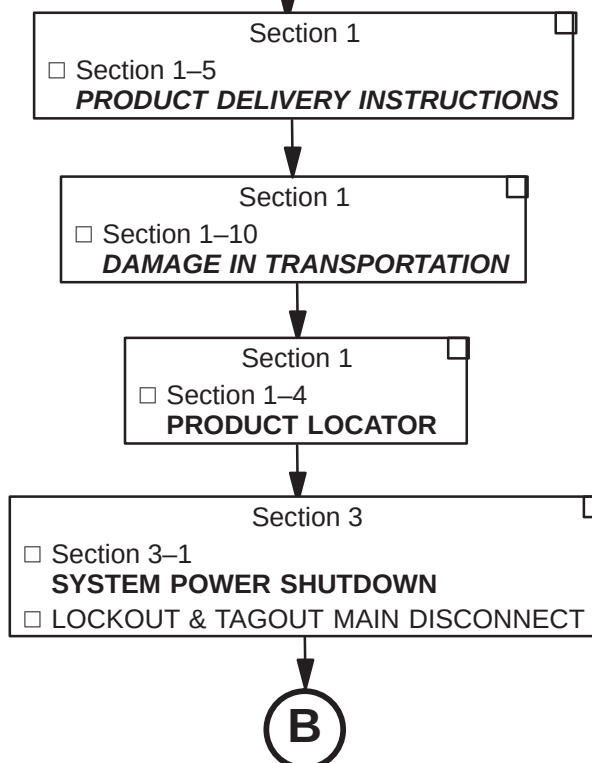
This upgrade flowchart has been developed assuming that all Signa HDx equipment has been delivered. **Make sure that every part required is available before starting.**

2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART

Procedures to be Completed by Field Engineer

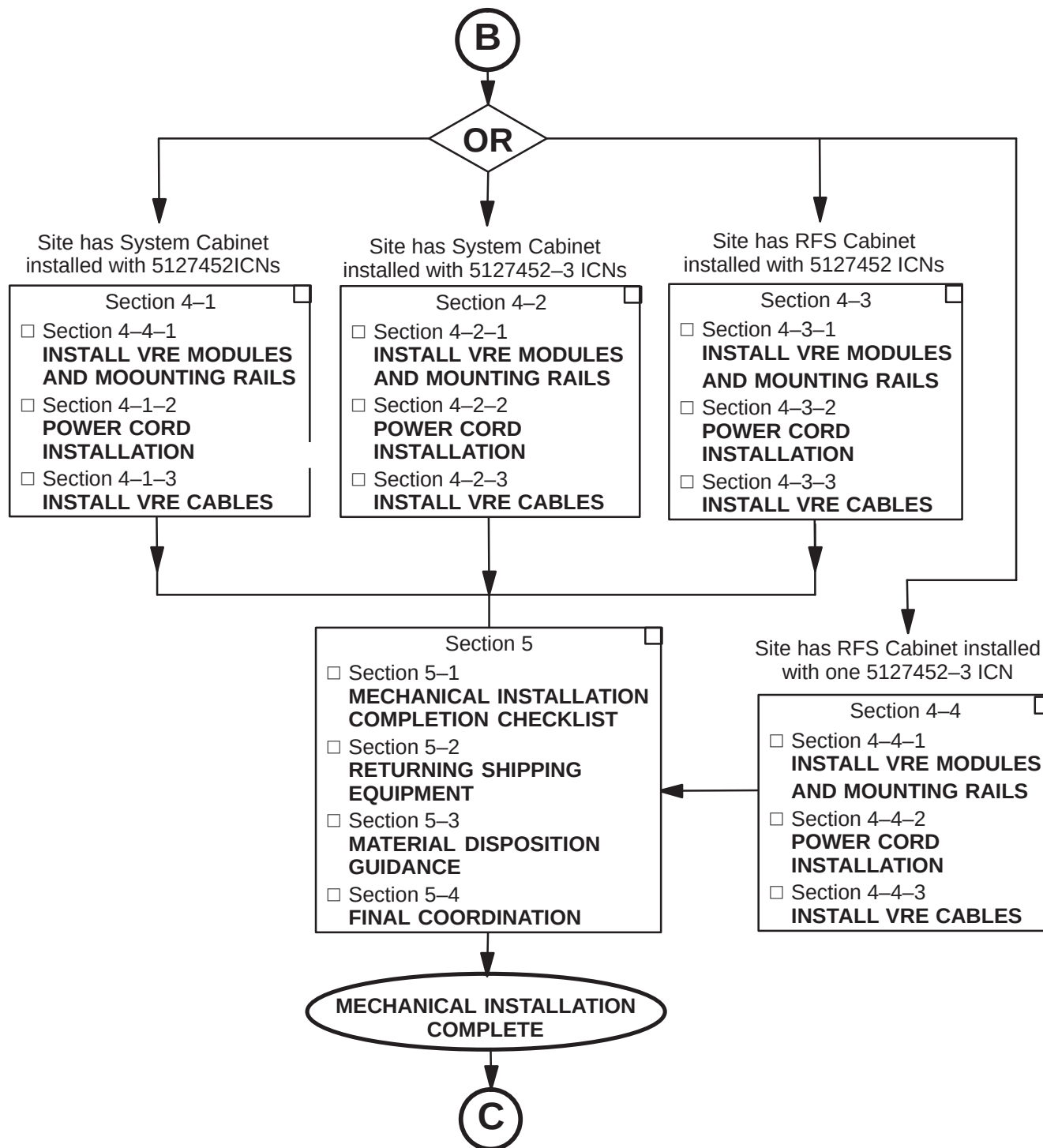


Procedures to be Completed by FE and/or Mechanical Installers



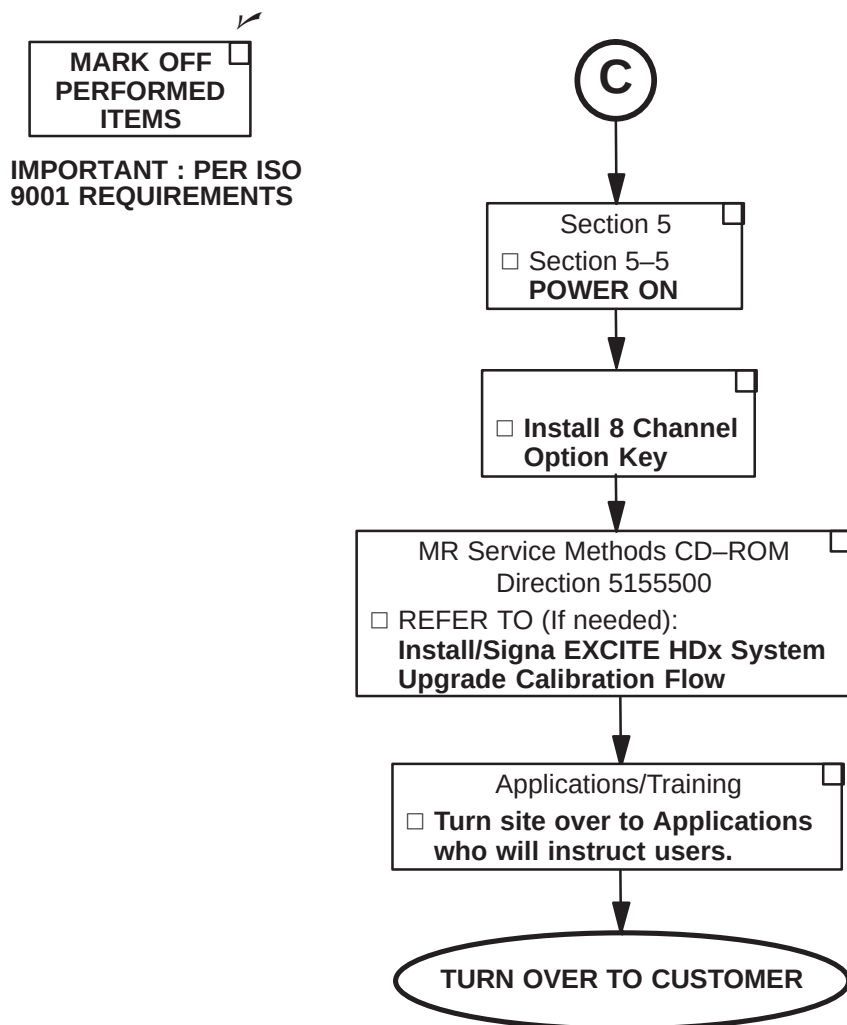
2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and/or Mechanical Installers



2-3 SYSTEM MECHANICAL UPGRADE FLOWCHART (Continued)

Procedures to be Completed by FE and Applications



SECTION 3 – POWER OFF

3-1 SYSTEM POWER SHUTDOWN



FATAL ELECTRIC SHOCK HAZARD!! LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU EVEN IF ALL PDU BREAKERS ARE OFF. TO PREVENT POSSIBLE FATAL ELECTRIC SHOCK VERIFY THAT FACILITY DISCONNECT IS OFF, LOCKED OUT, AND TAGGED BEFORE PROCEEDING. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE RESULT.

1. Make sure that site customer has archived all patient information. Execute "Save Info".
2. Verify that System performance baseline measurements have been obtained, especially:
 - SPT Full Test Mode
 - White Pixel Test
 - LV Shim
 - IIS
3. Record center frequency.
4. Ensure that Patient Transport is undocked with cradle in place and Head Coil carriage is at rear of travel.
5. Press the FULL OFF button on the PDU front control panel.
6. Notify field service and other installation personnel that are working at the site that the PDU main disconnect is to be shut off, locked out, and tagged.
7. Locate and shut off main disconnect supplying power to the PDU.
8. Perform lock out and tag out procedures at main disconnect to prevent inadvertent power restoration.
9. Verify absence of power in the PDU by checking for 0 volts on incoming power wiring

SECTION 4 – UPGRADE PROCEDURE

4-1 INSTALL M3335BH FOR EXISTING SYSTEM CABINET CONTAINING TWO 5127452 ICN'S

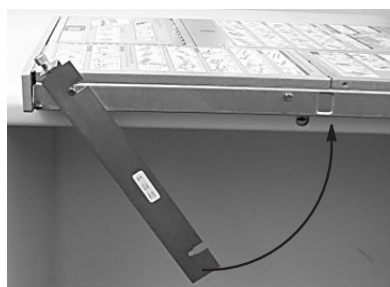
The following procedures provide upgrade information for upgrading a System Cabinet from 2 ICN (5127452) modules to 2 ICN (5127452-3) modules. The upgrade includes two ICN modules, ICN brackets, 6 outlet power strip (only needed for pre-HDx system cabinets), and power cables.

4-1-1 INSTALL VRE MODULES AND CABLES

- ① Detach all cables from back of ICN's.
- ② Remove ICN's (Remove (4) M5 screws per ICN, slide ICN modules out.)
- ③ Put 2 (5306485) brackets on each ICN as shown, no additional hardware is needed.

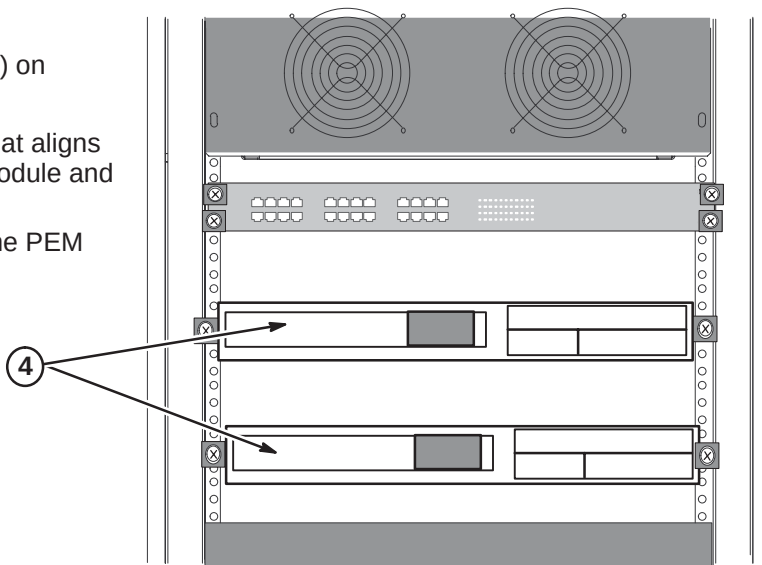


On the left side, the bracket is installed downward and snapped into place.



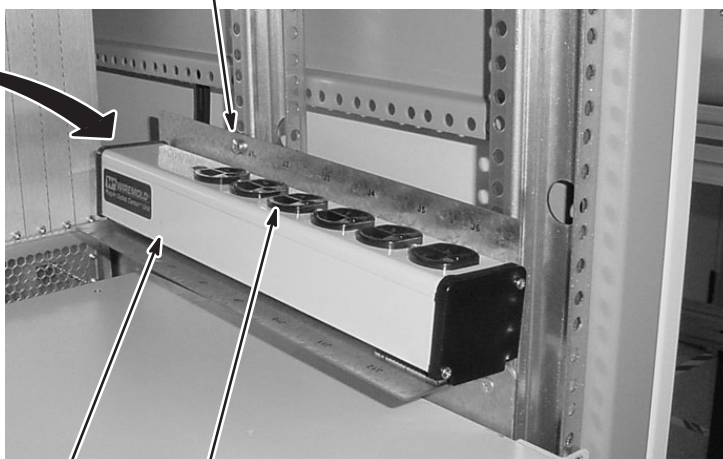
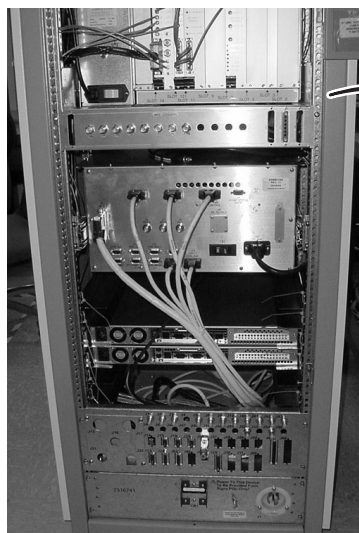
On the right side, the bracket must be installed upward at a 45 degree angle and then snapped into place.

- ④ Place the new ICN Modules (5127452-3) on the previously existing brackets.
- ⑤ Position Clip Nut over vertical rail hole that aligns with middle hole on each side of each module and fasten in place with supplied screw.
- ⑥ Secure modules in place by tightening the PEM screws on the ICN brackets.

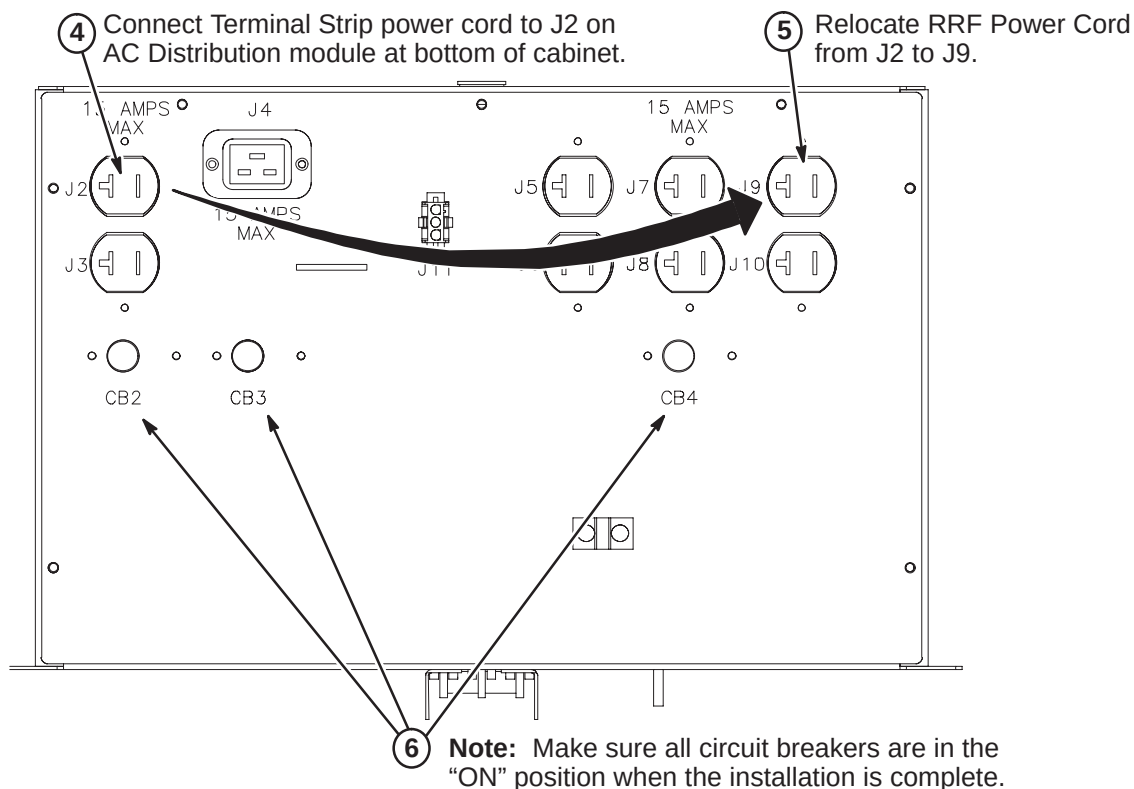


4-1-2 POWER CORD INSTALLATION

Note: Power strip is only needed for pre-HDx system cabinets.



- ① Install Power Strip Bracket (5150672) to vertical rails using Nut Clips (2180698-2) and M5 Screws (2279742-2) at this location at rear of cabinet.
- ② Attach Terminal Strip (5149828) to bracket using Dual Locking Tape (2341779 and 2341779-2).
- ③ Power cords connected to Terminal Strip:
J1, J6 – ICN 2 (5115594-3)
J4 – 24 Port Ethernet Switch (relocated from AC Distribution Module)



4-1-3 INSTALL VRE CABLES

Route and connect VRE Cables as shown below.

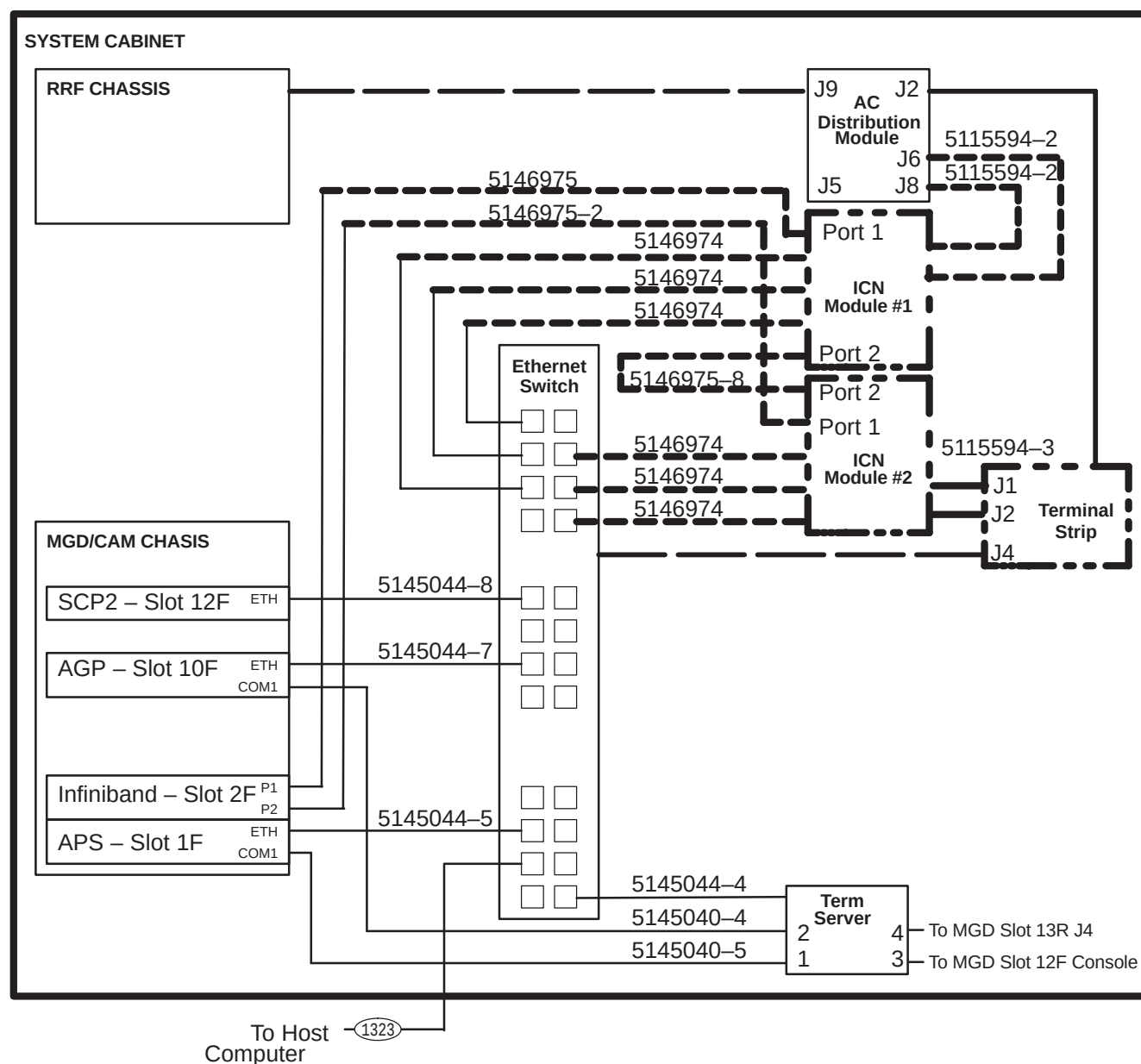
LEGEND:

NEW CABLE

REUSED AND RELOCATED CABLE

REPLACED MODULE

NEW MODULE



4-2 INSTALL M3335RC FOR EXISTING SYSTEM CABINET CONTAINING ONE 5127452-3 ICN.

The following procedures provide upgrade information for upgrading a System Cabinet from 1 ICN (5127452-3) module to 2 (5127452-3) ICN modules. The upgrade includes one new ICN module, ICN brackets, VRE cables, 6 outlet power strip and power cables.

4-2-1 INSTALL VRE MODULES AND MOUNTING RAILS

- ① Put two brackets on each ICN as shown, no additional hardware is required.

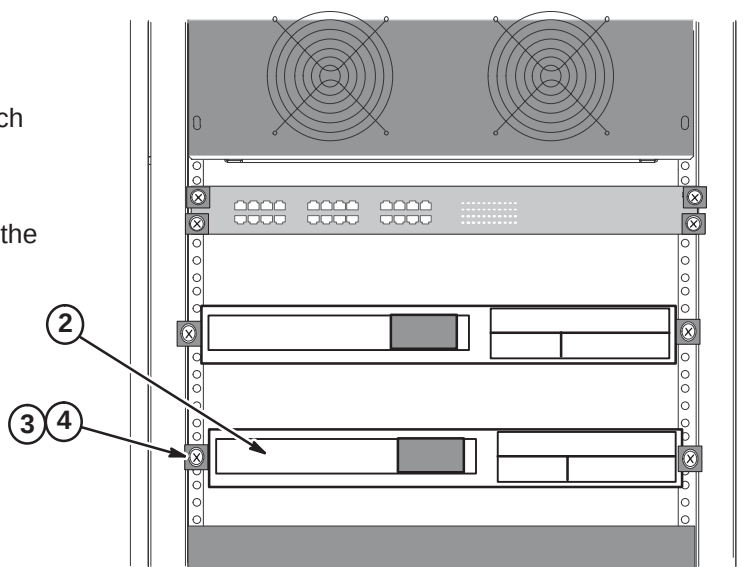


On the left side, the bracket is installed downward and snapped into place.



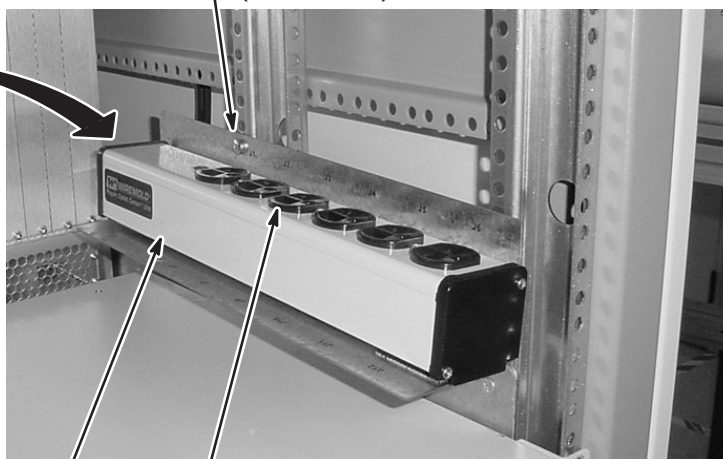
On the right side, the bracket must be installed upward at a 45 degree angle and then snapped into place.

- ② Place the new ICN Modules (5127452-3) on Mounting Rails.
- ③ Position clip nut over vertical rail hole that aligns with middle hole on each side of each module and fasten in place with supplied screw.
- ④ Secure ICN module in place by tightening the PEM screws on the ICN brackets.



4-2-2 POWER CORD INSTALLATION

Note: Power strip is only needed for pre- HDx system cabinets.



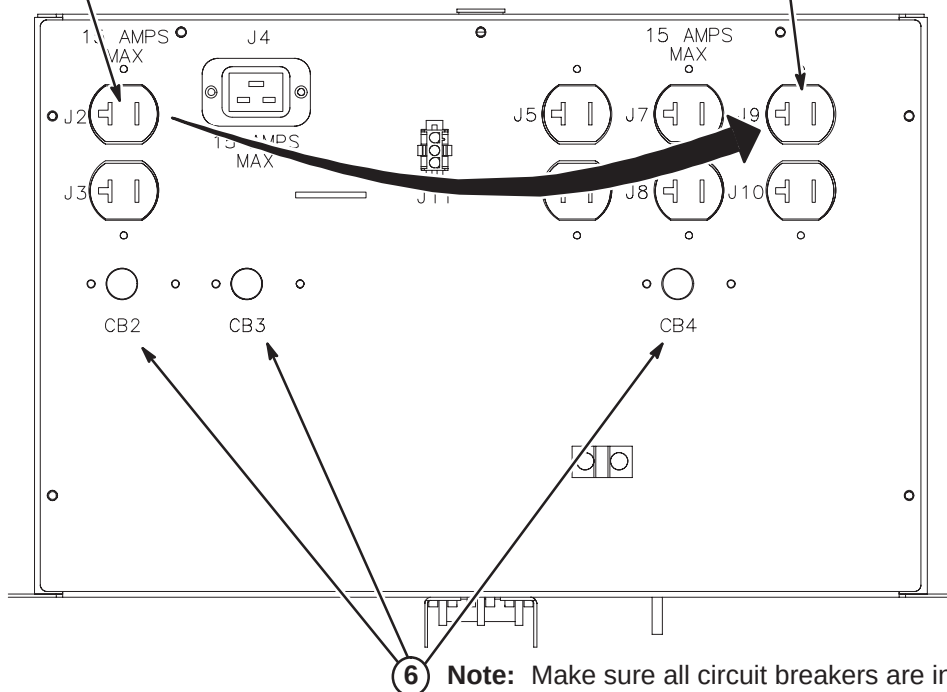
- ① Install Power Strip Bracket (5150672) to vertical rails using Nut Clips (2180698-2) and M5 Screws (2279742-2) at this location at rear of cabinet.

- ② Attach Terminal Strip (5149828) to bracket using Dual Locking Tape (2341779 and 2341779-2).

- ③ Power cords connected to Terminal Strip:
J1, J2 – ICN 2 (5115594-3)
J4 – 24 Port Ethernet Switch
(relocated from AC Distribution Module)

- ④ Connect Terminal Strip power cord to J2 on AC Distribution module at bottom of cabinet.

- ⑤ Relocate RRF Power Cord from J2 to J9.



4-2-3 INSTALL VRE CABLES

Route and connect VRE Cables as shown below.

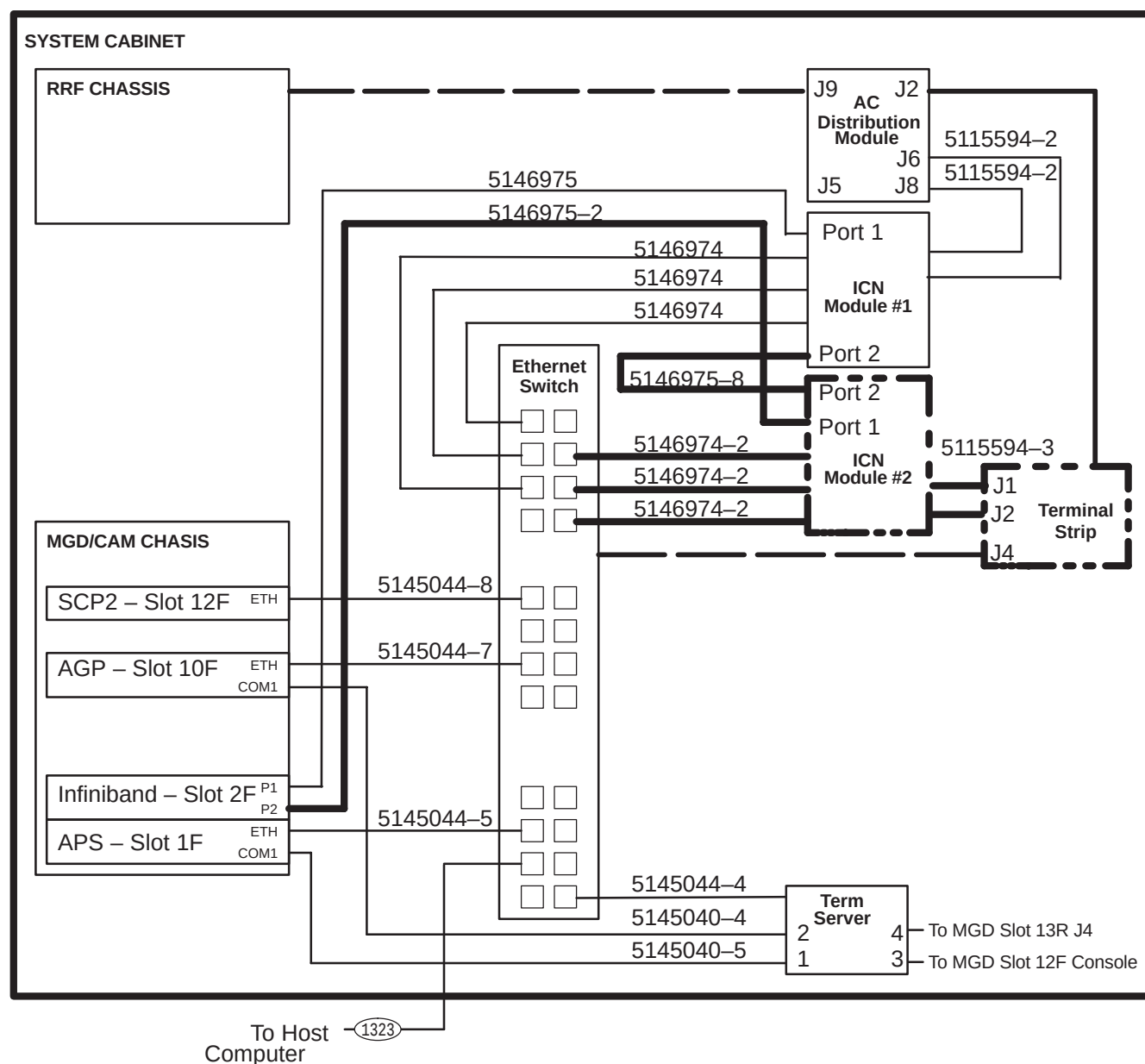
LEGEND:

NEW CABLE

REUSED AND RELOCATED CABLE

REPLACED MODULE

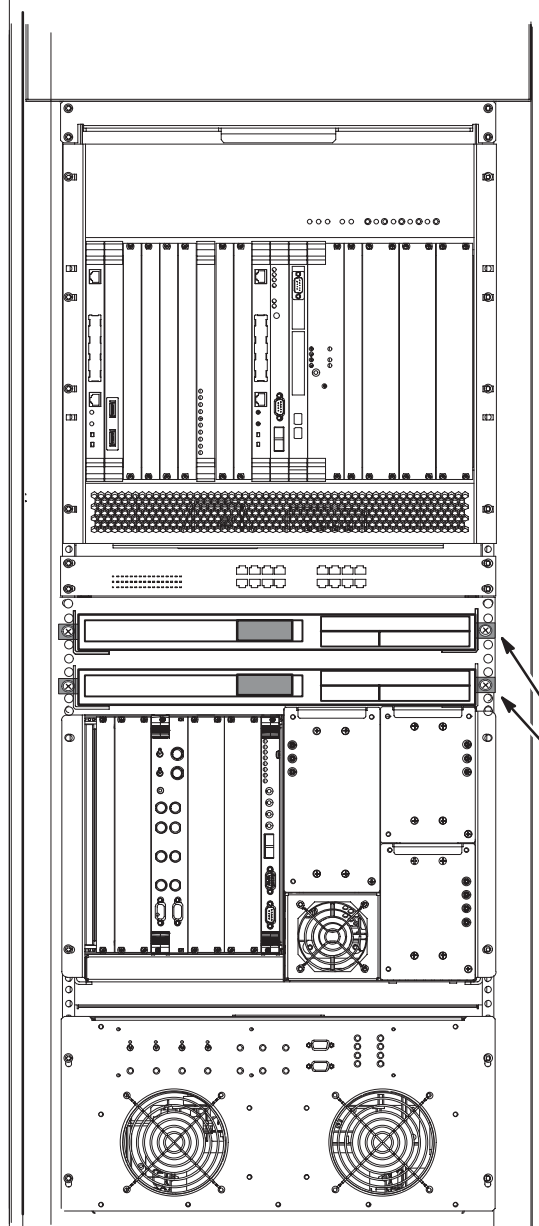
NEW MODULE



4-3 INSTALL M3335BH FOR RFS CABINET CONTAINING TWO 5127452 ICNS.

The following procedures provide upgrade information for upgrading a RFS Cabinet from 2 ICN (5127452) modules to 2 (5127452-3) ICN modules. The upgrade includes two new ICN modules, ICN brackets, and power cables.

4-3-1 INSTALL VRE MODULES AND MOUNTING RAILS

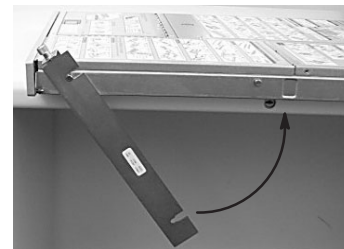


NOTE: If ICNs are installed at the bottom of the cabinet, follow the instructions in section 4-1.

- ① Detach all cables from back of ICNs.
- ② Remove ICNs (Remove (4) M5 screws per ICN, slide ICN modules out.)
- ③ Put two brackets on each ICN (5306485) as shown, no additional hardware is required.



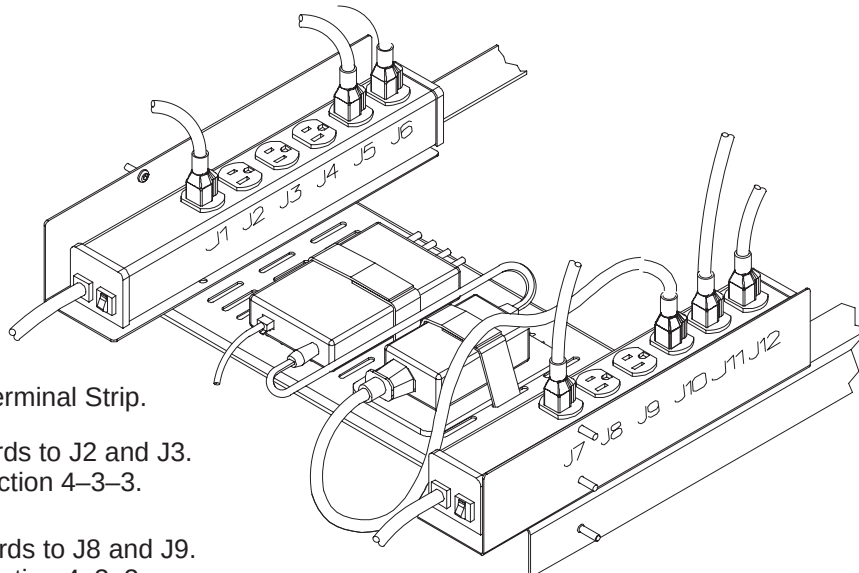
On the left side, the bracket is installed downward and snapped into place.



On the right side, the bracket must be installed upward at a 45 degree angle and then snapped into place.

- ④ Place the new ICN Modules (5127452-3) on the existing Mounting Rails.
- ⑤ Secure all modules in place by tightening the PEM screws on the ICN brackets.

4-3-2 POWER CORD INSTALLATION



- ① Route power cables to Terminal Strip.
- ② Connect ICN 1 power cords to J2 and J3. Refer to cable map in Section 4-3-3.
- ③ Connect ICN 2 power cords to J8 and J9. Refer to cable map in Section 4-3-3.
- ④ Coil up excess power cable and fasten to rear post with a tie wrap.

4-3-3 INSTALL VRE CABLES

Route and connect VRE Cables as shown below.

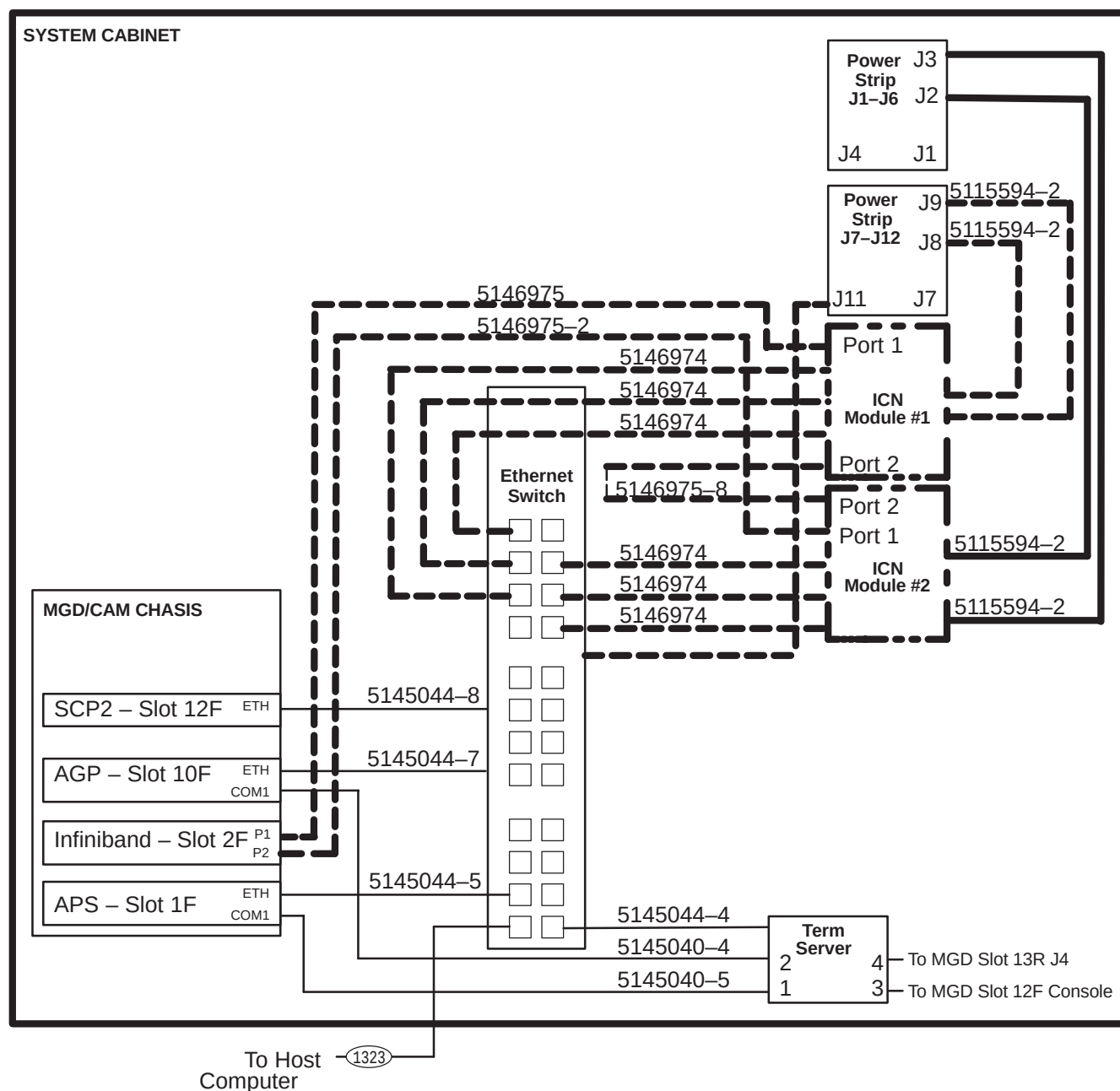
LEGEND:

NEW CABLE

REUSED AND RELOCATED CABLE

REPLACED MODULE

NEW MODULE



4-4 INSTALL M3335RC FOR RFS CABINET CONTAINING ONE 5127452-3 ICN.

The following procedures provide upgrade information for upgrading a RFS Cabinet from one (5127452-3) ICN module to two (5127452-3) ICN modules. This upgrade includes one new ICN module, ICN bracket, VRE cables and power cables.

4-4-1 INSTALL VRE MODULES AND MOUNTING RAILS

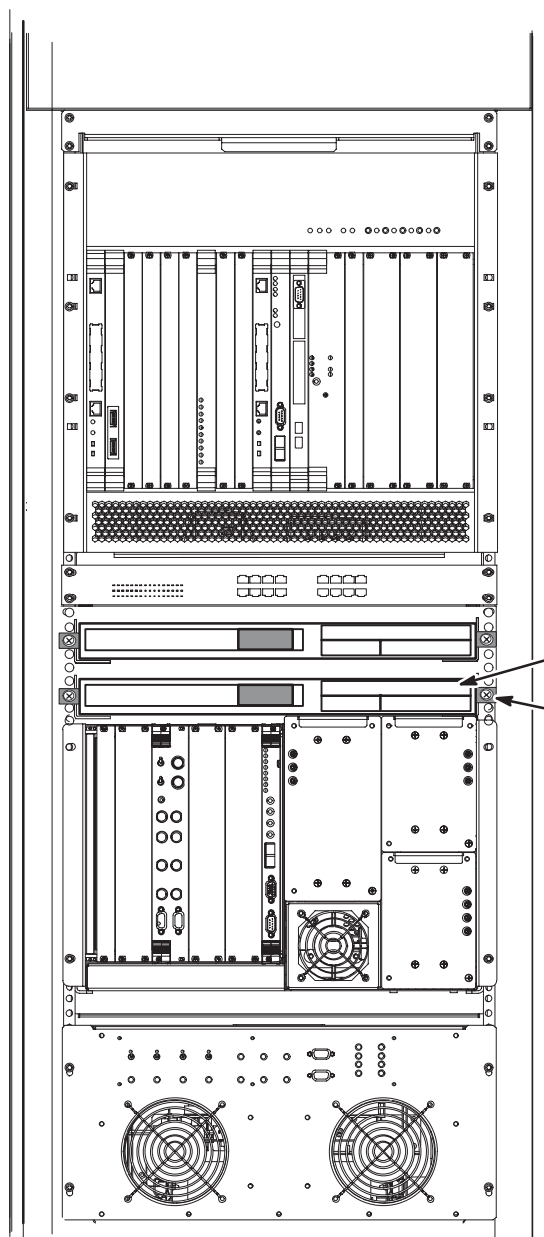
NOTE: If ICNs are installed at the bottom of the cabinet, follow the instructions in section 4-2.

- ① Put two brackets (5306485) on each ICN as shown, no additional hardware is required.

On the left side, the bracket is installed downward and snapped into place.

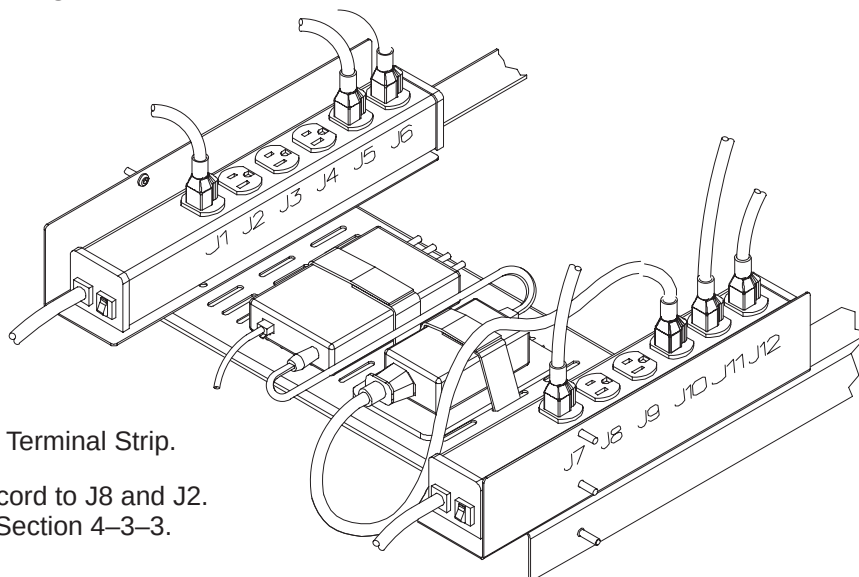


On the right side, the bracket must be installed upward at a 45 degree angle and then snapped into place.



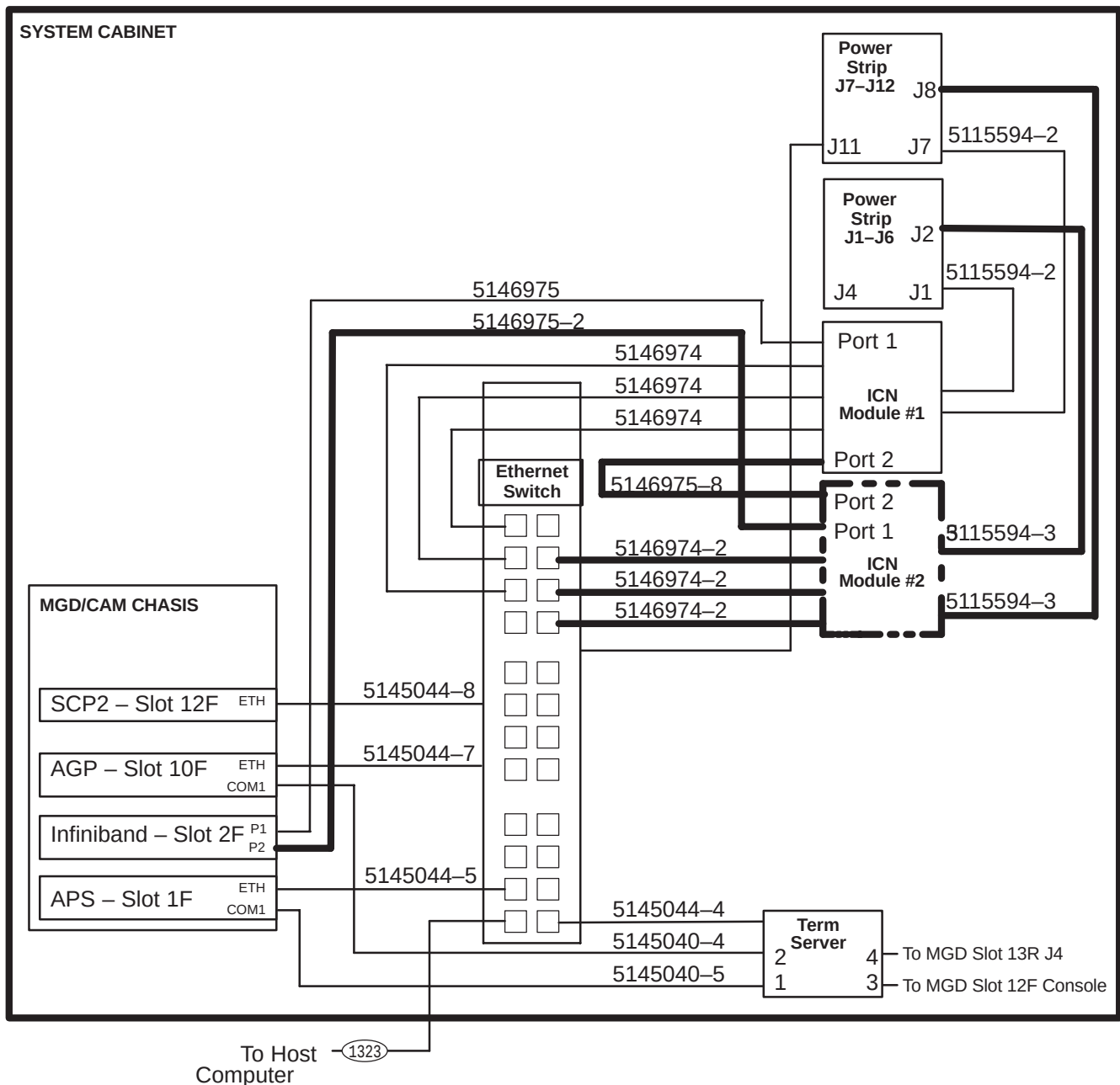
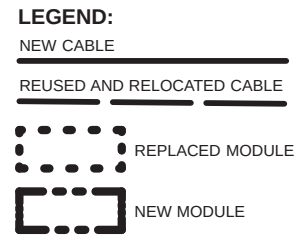
- ② Place the new ICN Module (5127452-3) on existing Mounting Rails.
- ③ Position clip nut over vertical rail hole that aligns with middle hole on each side of each module and fasten in place with supplied screw.
- ④ Secure ICN module in place by tightening the PEM screws on the ICN brackets.

4-4-2 POWER CORD INSTALLATION



- ① Route power cables to Terminal Strip.
- ② Connect ICN 2 power cord to J8 and J2. Refer to cable map in Section 4-3-3.

Route and connect VRE Cables as shown below.



SECTION 5 – COMPLETION

5-1 MECHANICAL INSTALLATION COMPLETION

The following steps need to be reviewed and completed by GE Field Engineer, especially if the system was installed by non-GE personnel. Check off boxes at the front of each item below when complete.

- ☐ Check mechanical flowchart to make sure all steps are complete (Section 2)
- ☐ Resolve shipment shortages
- ☐ Resolve omissions by mechanical contractors
- ☐ Make sure all rating plates are installed
- ☐ Recheck all wiring connections using cable map (Section 4)
- ☐ Correct for wiring errors if necessary
- ☐ Complete Sections 5-2, 5-3, and 5-4.

5-2 RETURNING SHIPPING EQUIPMENT

The shipping material must be returned **as soon as possible**. Prompt return is required so that production shipments can continue with these labor saving dollies. These can be included with the obsolete equipment and materials being returned in Section 5-3. Recycling Operation will route them to appropriate location.

5-3 MATERIAL DISPOSITION GUIDANCE

Refer to Section 1-3. The equipment or material being disposed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

5-4 FINAL COORDINATION

1. Replace cabinet covers that have not been previously replaced.
2. Dock Patient Transport in position at front of bore.
3. Record and enter applicable data into applicable site configuration files and records.
4. Complete Product Locator information. See Section 1-4.

Note

Failure to fill out and return Product Locator Cards may result in failure of your site to receive future FMI's.

5. Leave the site's set of Manuals and CD-ROM at site. Organize and set up reference cabinet for the Manuals.
6. Locate any new Material Safety Data Sheets (MSDS). They must be retained on site. Customer is to be informed that material with MSDS information was brought into site and customer should know/decide where MSDS be retained.
7. Verify that coordination of application support for instruction of site users on Software Release 11.x has been made before returning system to the users. Turn site over to applications who will instruct users.

5-5 POWER ON

1. Notify field service and other installation personnel that are working at the site that the PDU main disconnect locks and tags will be removed so PDU power can be turned on.
2. Restore power to the PDU from main disconnect.
3. Refer to the CD-ROM, Installation Calibration Flow, and perform the following:
 - PDU Voltage Checks Compact > PDU E-Stop Check
 - Startup Checks Compact
 - Transformer Taps Compact
4. Press the FULL ON button on the PDU front control panel.
5. Except for the HFD Gradient power modules, switch all cabinet circuit breakers to ON. The HFD Gradient power module circuit breakers should remain OFF for one hour after HVAC power up to allow room humidity to stabilize.
6. **Make sure the UPS circuit breaker is switched to the ON position.** This is required to feed power to the Operator Workspace.